METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET C-201

C-201 · Loop circulator

DOCUMENT NO.	:	MEOH-V1-ME-DS-007
DOCUMENT TITLE	:	Datasheet C-201
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

C-201 · Loop circulator

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-002.

General		
Service	Recirculates the loop gas (make-up joins the suction)	MEOH-PID-002
Type	Single-stage centrifugal, barrel casing, variable-speed motor (FIC-201 → speed)	P&ID note 4
Seals	Dry gas seals (vendor)	typical
Operating (model, design point)		
Flow	1928 kNm³/h = 30051 m³/h at suction	FIC-201
Gas M / cp	6.57 kg/kmol / 31.7 kJ/kmol K	stream after the make-up tie-in
Suction	75 bar g / 40 °C	
Discharge	80.6 bar g / 48.6 °C	pressure ratio 1.074
ΔT / head	8.6 K / ≈ 29.8 kJ/kg	η 0.72, (k−1)/k 0.275 (model)
Shaft power	6.53 MW at 80 % speed	P.kW, JI-201
Capacity		
Flow at 100 % speed (75 bar g)	2410 kNm³/h	K.F100
Power at 100 % speed	≈ 12.6 MW	model formula
Motor rating	per electrical load list MEOH-V1-EE-LL-001	
Settle-out	loop pressure, both sides equalise (≤ zone A design)	
Design		
Casing design pressure / temperature	100 bar g / 120 °C	Zone B; discharge design T
Casing material	Forged carbon / low-alloy steel (vendor)	EQL §3
Code	API 617 (assumed)	
Notes		
1	FSLL-201 (I-2) and LSHH-204 (I-3, also stops C-201).	MEOH-PID-002
2	No high-temperature stop on C-201 for total cooling-water loss (EQ-em-04).	